MIL-S-18538(NOrd)

5 May 1955

SUPERSEDING
NAVORD OS 3799 H
20 APRIL 1954

MILITARY SPECIFICATION

SIGNAL, MINIATURE PRACTICE BOMB, MK 4 MOD 3 AND MOD 4

1. SCOPE

1.1 This specification governs the manufacture, assembly, and preparation for delivery of Miniature Practice Bomb Signal Mk 4 Mod 3 and Mk 4 Mod 4, and the methods of inspection and tests upon which acceptance will be based. The signal is used in practice bombs to produce smoke when the bomb hits a hard surface.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, drawings, and publications, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

UU-T-111	-- Tape; Paper, Gummed (Sealing and Securing).
PPP-B-601	-- Boxes, Wood, Cleated-Plywood.

MILITARY

JAN-P-106	-- Packaging and Packing for Overseas Shipment -- Boxes; Wood, Nailed.
JAN-P-125	-- Packaging and Packing for Overseas Shipment -- Barrier -- Materials, Waterproof, Flexible.
JAN-P-133	-- Packaging and Packing for Overseas Shipment -- Boxes, Set-Up, Paperboard.

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DRAWINGS

BUREAU OF ORDNANCE

LD 90683	-- Miniature Practice Bomb Mk 23,
LD 166560	-- Miniature Practice Bomb Signal Mk 4 Mod 3,
LD 256093	-- Miniature Practice Bomb Signal Mk 4 Mod 4,
Dwg 398978	-- Primer Sensitivity Testing Fixture Mk 135 Mod 0, and all documents listed thereon.

STANDARDS

MIL-STD-105	-- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-- Marking for Shipment and Storage.

PUBLICATIONS

BUREAU OF ORDNANCE

OP 400	-- General Instructions for the Design, Manufacture, and Inspection of Naval Ordnance Equipment.
NavOrd OD 5823	-- Sensitivity Test of Primers and Detonators.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. -- The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

CODE OF FEDERAL REGULATIONS

49 CFR 71-78	-- Transportation, Interstate Commerce Commission, Explosives and Other Dangerous Articles.

(The Interstate Commerce Commission regulations are now a part of the Code of Federal Regulations (1949 Edition-Revised 1950) available from the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Orders for the above publication should cite "49 CFR 71-78 (Rev. 1950).")

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3. REQUIREMENTS

3.1 General.

3.1.1 The Miniature Practice Bomb Signal Mk 4 Mod 3 and Mod 4 covered by this specification shall be manufactured in accordance with Bureau of Ordnance LD 166560 or LD 256093, as applicable, and all documents listed thereon.

3.1.2 The requirements and procedures established in OP 400 shall be adhered to wherever applicable.

3.2 Primer Mk 119 Mod 0.

3.2.1 Priming mixture. -- The priming mixture specified by the drawing shall be a thoroughly mixed homogeneous composition as determined by visual examination.

3.2.2 Sensitivity. -- The sensitivity of the Primer Mk 119 Mod 0 shall be such that the drop height of a two ounce steel ball required to produce 50 percent firing plus five times the standard deviation shall not exceed 18.0 inches. The methods of evaluating sensitivity shall be in accordance with 4.5.1.

3.2.3 Storage. -- The primers and primed shells shall be stored in dry airtight containers until used. The temperature of the storage space shall not exceed a temperature of 100 degrees F.

3.3 Signal functioning. -- The signal after assembly in the Miniature Practice Bomb Mk 23 shall function on water and land impacts to produce a visible smoke signal. Testing and acceptance shall be in accordance with 4.5.2.

3.4 Pyrotechnic and expellent charge. -- The pyrotechnic and expellant charge shall be stored in dry airtight containers until used. The storage space shall not exceed a temperature of 100 degrees F.

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4. QUALITY ASSURANCE PROVISIONS AND TEST REQUIREMENTS

4.1 Sampling. -- Unless otherwise specified, and when applicable, the sampling plans and procedures used by the Government Inspector in the determination of the acceptability of products submitted by a supplier for Government Inspection will be in accordance with the provisions of Military Standard MIL-STD-105, "Sampling Procedures and Tables for Inspection by Attributes".

4.1.1 Lot. -- As applied to the Government Inspection of units of product, the term "Lot" shall mean "Inspection Lot", i.e., a collection of units of the product used as a basis for Government Inspection. Unless otherwise specified, the number of units of product in "Inspection Lots" shall be as determined by the Government Inspector, and may differ from the quantity designated in the contract or order as a lot for production, shipment or other purpose.

4.2 Preproduction sample. -- Before entering into quantity production, a preproduction sample of 200 Miniature Practice Bomb Signals Mk 5 (designated modification) shall be delivered by the Contractor for a test to an agency designated by the Procuring Agency. These signals shall be manufactured by the same procedures and processes and at the same location proposed by the Contractor for the execution of the contract. When a preproduction sample is received which proves to be satisfactory for the purpose intended, the Contractor will be notified and will be authorized by the Procuring Agency to proceed with the submission of the product for acceptance. Accepted samples will become the property of the Procuring Agency and will be included in the number called for in the contract schedule.

4.2.1 Preproduction sample for a subsequent contract. -- The necessity for a preproduction sample will be determined by the Procuring Agency when production under a new contract by the same Contractor at the same location follows the manufacture of satisfactory material as covered by this specification.

4.3 Production lot. -- A production lot shall consist of 10,000 signals plus such excess as is required for tests. The production lot should not be confused with the lot for inspection as defined in MIL-STD-105.

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4.4 Inspection. -- The general inspection procedures shall be in accordance with "General Instructions for the Design, Manufacture, and Inspection of Naval Ordnance Equipment", OP 400; Government Inspection shall be in accordance with the provision of general procedure for Government Inspection and Contractor's Inspection shall be in accordance with Inspection by Contractor, as contained in OP 400.

4.5 Test procedures.

4.5.1 Sensitivity test of primer. -- The sensitivity test shall be conducted on each lot of 10,000 primers (maximum) using Test Set (Primer) Mk 135 Mod 0 equipped with a two ounce steel drop ball, a firing pin similar to 553214 Pc 5 but which has a 0.075 inch radius on the point, and suitable primer holders. The sensitivity of the primers shall be determined by the "Run Down Method" with 50 primers tested at each one half inch interval of height. Installation, adjustment and operation of Test Set Mk 135 Mod 0 and "Run Down Method" of determining sensitivity are described in NavOrd OD 5823. Frankford Table, PRNC-NOL 5242 (11-46), is a convenient form for recording sensitivity observations and calculations. Disposition of the lot of primers shall be in accordance with the following formula:

$\bar{h} + 5s$ shall be less than 18.00 inches for lot acceptance.

4.5.2 Functioning test. -- Fifty-six signals chosen at random from each lot of 10,000 signals (maximum) shall be tested for functioning by installation in Miniature Practice Bombs Mk 23. The bombs shall be released in groups of four from an airplane, travelling at an altitude of 3000 feet and ground speed of 150 knots, so as to cause impact alternately on water and land targets. A bomb signal which does not produce a visible smoke signal when observed at an altitude of 3000 feet shall be classed as a defective.

4.5.3 Acceptability. -- Acceptance of the lot shall be based on Table I. Only the minimum number of bomb signals required to determine acceptance under this plan shall be used.

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Table I - Functioning Test

Sampling Plan (.05/.05 : .20/.10)			
When this number of defectives is found	The lot will be rejected if this number or less have been tested	Testing will be continued if this number has been tested	The lot will be accepted if this number has been tested
0	---	0-15	16
1	---	1-23	24
2	---	2-31	32
3	16	17-39	40
4	24	25-39	40
5	32	33-47	48
6	40	41-55	56
7	56	-----	--

5. PREPARATION FOR DELIVERY

5.1 Packaging (domestic shipment of inert cartridges). -- Twenty-five cartridges shall be packed in paperboard set-up boxes, full telescope type, conforming to the requirements of Military Specification JAN-P-133. Each box shall have suitable dividers to prevent damage to the cartridges during handling and shipment.

5.2 Packaging (domestic and overseas shipment of loaded cartridges). -- Loaded signals shall be repacked in the boxes specified in 5.1, and each box shall be sealed with 1 1/2 inch kraft paper tape. The packaging shall conform to the requirements of Federal Specification UU-T-111 and the Code of Federal Regulations 49 CFR 71-78.

5.3 Packing (domestic shipment of inert cartridges). -- Twenty boxes of inert cartridges specified in 5.1 shall be packed in cleated plywood or nailed wood boxes conforming to the requirements of Specifications PPP-B-601 or JAN-P-106, respectively.

5.4 Packing (domestic and overseas shipment of loaded cartridges). -- The cartridges shall be packed as specified in 5.3, except each wooden box shall be lined with type L2 or M barrier material conforming to the requirements of Military Specification JAN-P-125. In addition, all containers shall conform to the requirements of the Code of Federal Regulations 49 CFR 71-78.

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5.5 Marking. -- In addition to any special marking required by the contract or order, all containers shall be marked as specified in 5.5.1 and in compliance with Military Standard MIL-STD-129 and the Code of Federal Regulations 49 CFR 71-78.

5.5.1 The following information shall be stenciled on one side and one end of all containers.

(insert quantity)

U.S. Navy
Practice Bomb Signals
Mk 4 mod ____

Mfg Lot Mon Year

6. NOTES. -- There are none applicable.

Patent notice. -- When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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Commanding Officer
Naval Air Engineering Center
Engineering Specifications and Standards
Department (ESSD), Code 93
Lakehurst, NJ 08733

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

MIL-S-18538(NORD)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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